

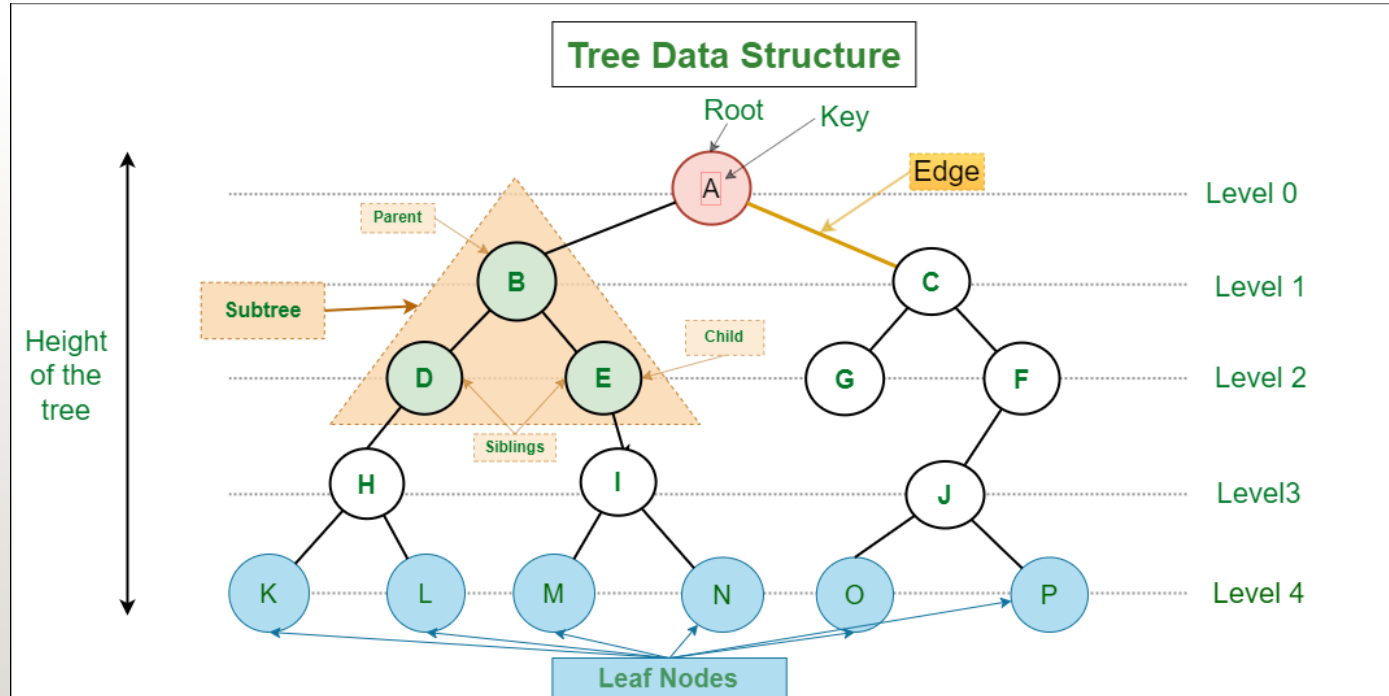
Decision Trees

ARIES Weeks

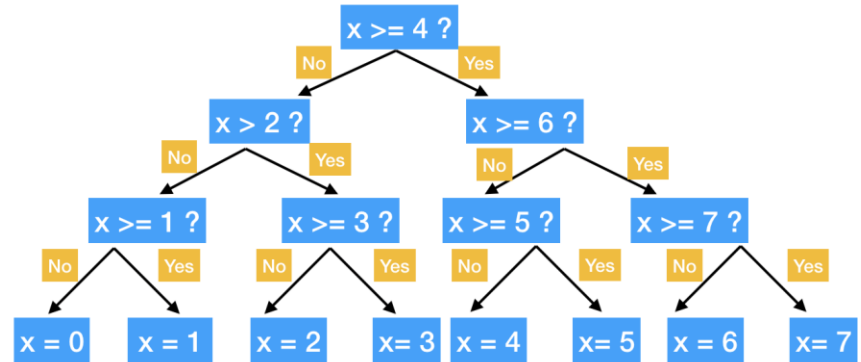
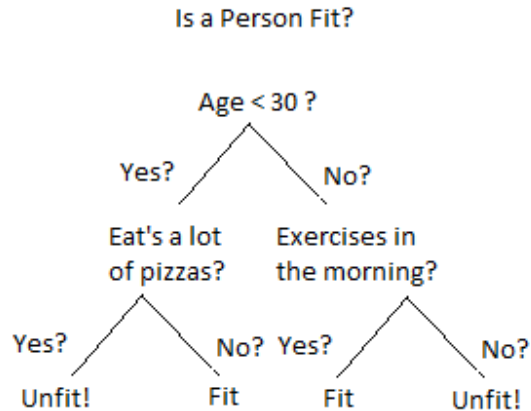
CONTENTS

- Introduction to decision trees(what are they)
- Importance of trees
- Difference between classification trees and regression trees
- Splitting Criteria(Gini Impurity and MSE Calculation)
- Algorithm for construction of trees
- Overfitting(and how to prevent it)
- Pruning(Weakest Link Pruning)

Tree Data Structure



Decision Trees



IMPORTANCE OF DECISION TREES

- Interpretable
- Require minimal data preprocessing (handle both numerical and categorical features, don't need normalization)
- Automatically perform feature selection by choosing the most informative splits
- Model non-linear relationships and interactions between features without explicit feature engineering.
- They form the building blocks of powerful ensemble methods like Random Forests and Gradient Boosting, which consistently achieve state-of-the-art performance across diverse tasks.

Two Flavors of Trees

CLASSIFICATION

Will customer churn?

YES

NO

REGRESSION

What's the house price?

\$450,000

Same algorithm, different output

Why Trees Work

Classification: Linear vs Tree

Linear Model: **65% accuracy**

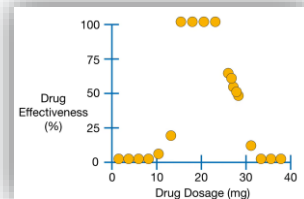
Tree Model: **95% accuracy**

Loves Popcorn	Loves Soda	Age	Loves Cool As Ice
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

Regression: Continuous Predictions

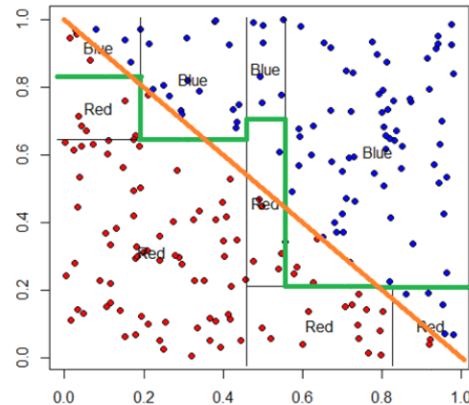
Linear regression: straight line

Tree: step function (better fit)



Trees create rectangular decision boundaries

They partition the feature space into regions



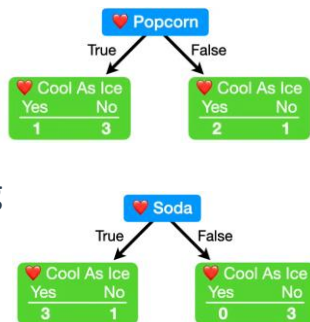
Splitting Criteria Comparison

CLASSIFICATION: Gini Impurity

$$Gini = 1 - \sum (p_i)^2$$

Goal: Minimize class mixing

Pure node $\rightarrow$ Gini = 0

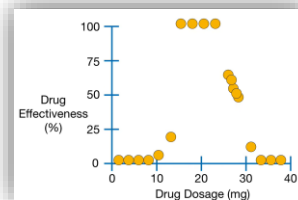


REGRESSION: Mean Squared Error

$$MSE = (1/n) \sum (y_i - \bar{y})^2$$

Goal: Minimize variance

Tight cluster $\rightarrow$ MSE = 0

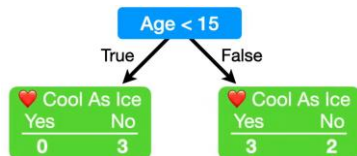


Both measure: How messy is this node?

Splitting Criteria Comparison

CLASSIFICATION: Gini Impurity

$$Gini = 1 - \sum (p_i)^2$$



Goal: Minimize class mixing

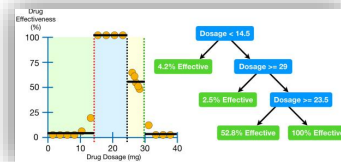
Pure node → Gini = 0

REGRESSION: Mean Squared Error

$$MSE = (1/n) \sum (y_i - \bar{y})^2$$

Goal: Minimize variance

Tight cluster → MSE = 0



Both measure: How messy is this node?

What Makes a Good Split?

BAD SPLIT (Mixed)

Left node: 3 Yes, 4 No

Right node: 2 Yes, 3 No

Gini = 0.48

GOOD SPLIT (Pure)

Left node: 0 Yes, 7 No

Right node: 5 Yes, 0 No

Gini = 0.00

Lower Gini = Purer nodes = Better split

Our Playground: Two Datasets

CLASSIFICATION: Loan Approval

Income	Age	Credit	Approved
50k	25	700	No
80k	35	750	Yes
45k	28	680	No
90k	40	800	Yes
60k	30	720	Yes
40k	22	650	No
95k	45	820	Yes
55k	32	710	Yes

REGRESSION: House Price

Size	Beds	Location	Price
1200	2	Urban	300k
1800	3	Suburban	450k
1000	2	Rural	250k
2200	4	Urban	550k
1500	3	Suburban	400k
900	1	Urban	280k
2500	4	Suburban	600k
1300	2	Rural	320k

We'll focus on classification, then show regression parallel

The Algorithm: Recursive Splitting

```
def build_tree(data, target_type, depth=0):  
    if stopping_criterion(data, depth):  
        if target_type == "classification":  
            return Leaf(majority_class(data))  
        else: # regression  
            return Leaf(mean(data.target))  
  
    best_feature, best_threshold = find_best_split(  
        data,  
        criterion="gini" if target_type=="classification" else "mse"  
    )  
    left_data, right_data = split(data, best_feature, best_threshold)  
  
    left_tree = build_tree(left_data, depth+1)  
    right_tree = build_tree(right_data, depth+1)  
  
    return Node(best_feature, best_threshold, left_tree, right_tree)
```

Three steps: find best split → split data → recurse

Let's Calculate: Income > 70k?

Split on Income $\leq$ 70k:

Left (Income $\leq$ 70k): 50k $\rightarrow$ No, 45k $\rightarrow$ No, 60k $\rightarrow$ Yes, 40k $\rightarrow$ No, 55k $\rightarrow$ Yes

$\rightarrow$ 2 Yes, 3 No

$$\rightarrow \text{Gini} = 1 - (2/5)^2 - (3/5)^2 = 1 - 0.16 - 0.36 = 0.48$$

Right (Income > 70k): 80k $\rightarrow$ Yes, 90k $\rightarrow$ Yes, 95k $\rightarrow$ Yes

$\rightarrow$ 3 Yes, 0 No

$$\rightarrow \text{Gini} = 1 - (3/3)^2 - (0/3)^2 = 0.00$$

$$\text{Weighted Gini} = (5/8) \times 0.48 + (3/8) \times 0.00 = 0.30$$

This is what the algorithm does at every node!

Your Turn!

Calculate: Split on Age > 30?

60 seconds - calculate the weighted Gini

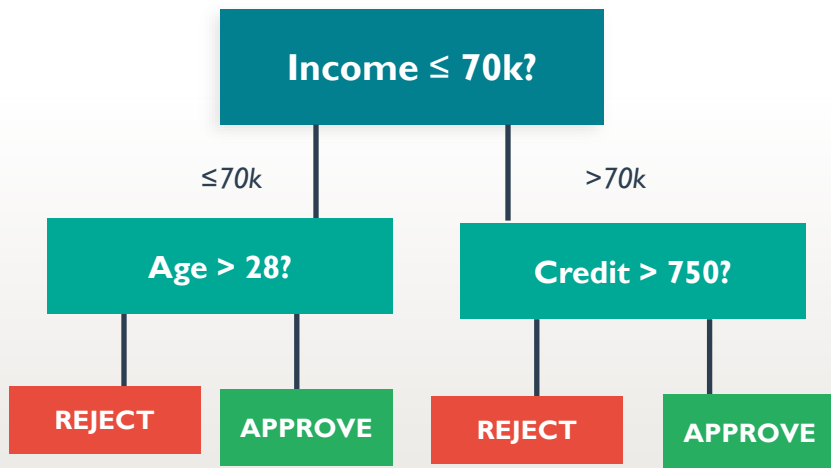
Comparing Splits

Split Criterion	Weighted Gini
Income > 70k	0.30
Age > 30	0.38
Credit > 720	0.31

Income > 70k wins! (Lowest Gini)

That's our root node split

Building the Tree (Recursively)



Same process recursively until nodes are pure or we hit stopping criteria

When to Stop Splitting?

- ✓ All samples in node have same class (pure node)
- ✓ Maximum depth reached (e.g., depth = 5)
- ✓ Too few samples to split (e.g., < 5 samples)
- ✓ No improvement in Gini/MSE from splitting

Otherwise you OVERFIT → Speaking of which...

Regression Trees: Quick Parallel

Split on Size ≤ 1500 ?

Left (Size ≤ 1500): [300k, 320k, 280k, 250k]

→ mean = 287.5k

→ MSE = $(1/4)[(300-287.5)^2 + (320-287.5)^2 + (280-287.5)^2 + (250-287.5)^2]$

→ MSE = 656.25

Right (Size > 1500): [450k, 550k, 400k, 600k]

→ mean = 500k

→ MSE = $(1/4)[(450-500)^2 + (550-500)^2 + (400-500)^2 + (600-500)^2]$

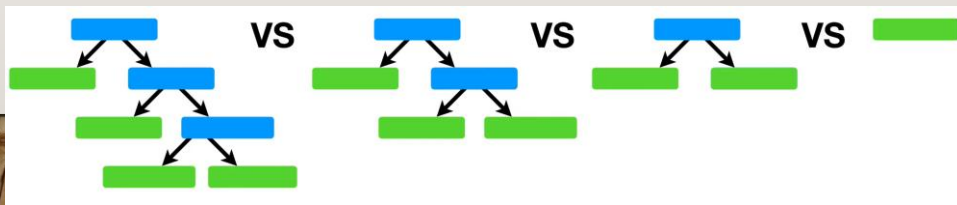
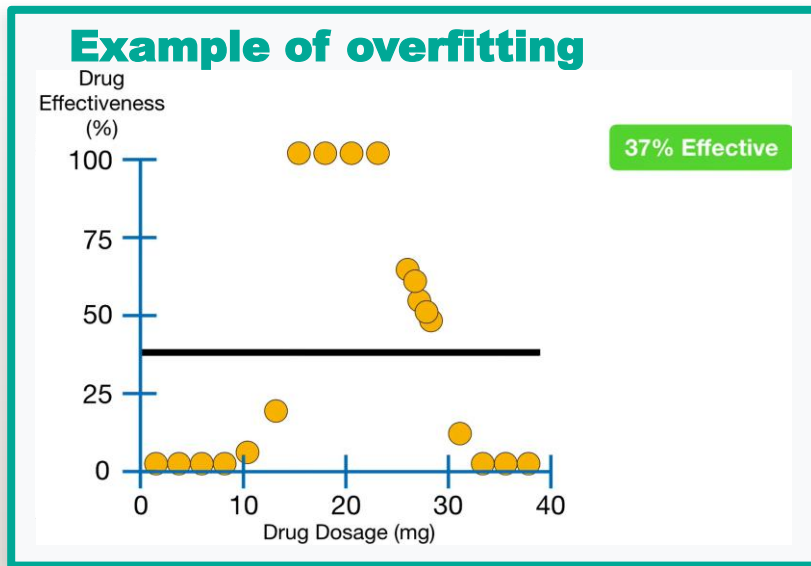
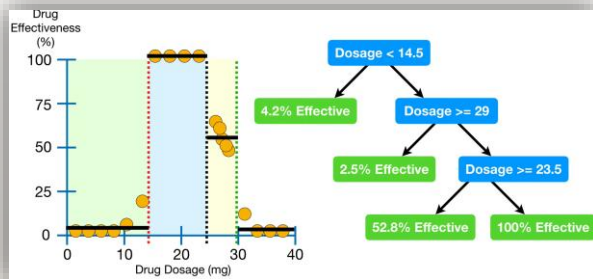
→ MSE = 6250

Weighted MSE = $(4/8) \times 656.25 + (4/8) \times 6250 = 3453$

Compare all features → pick lowest MSE → that's the split!

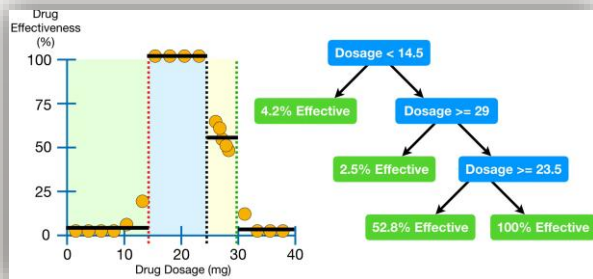
Overfitting in Regression Trees

Other than the methods used in classification trees, another method used to prevent overfitting in regression trees is called **pruning**

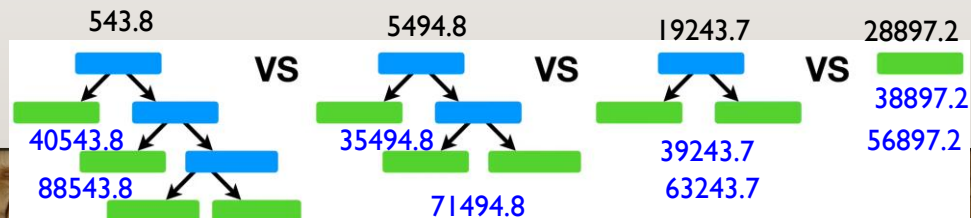


Weakest Link Pruning

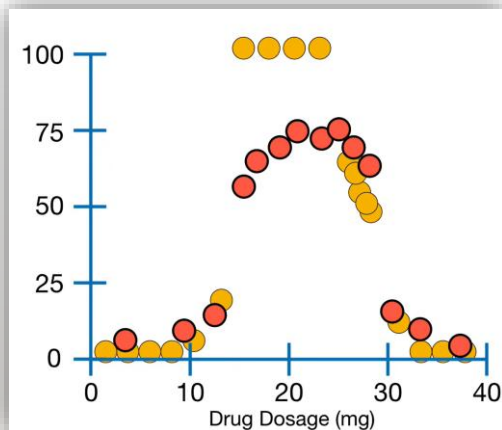
Tree score = $MSE + \alpha T$
 where T (Tree Complexity Penalty) =
 L (L = No. of Leaf Nodes)
 α is a tuning parameter, decided using
 Cross-Validation



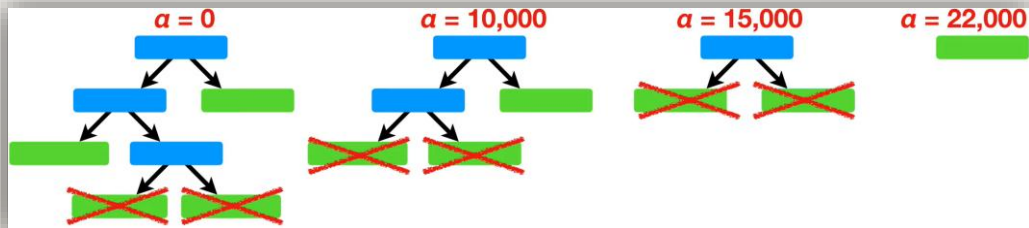
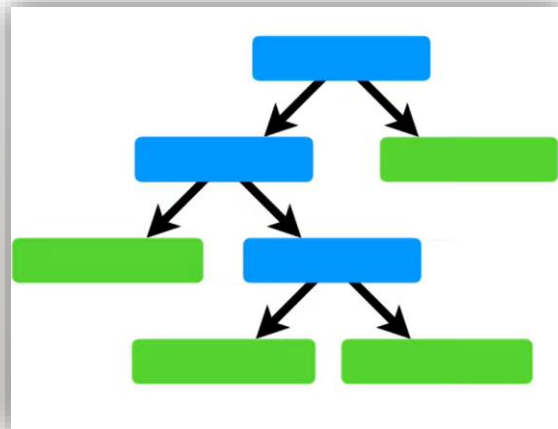
Weakest link pruning works by calculating a tree score for each viable tree and then choosing the tree with the minimum tree score.



Finding α



Use all of the data and create a full-sized regression tree
Then keep increasing the value of α until pruning gives us a better tree score



What Do Leaf Nodes Predict?

CLASSIFICATION

Leaf Node

7 Yes

2 No

→ Predict "Yes"

REGRESSION

Leaf Node

[450k, 480k,
490k]

→ Predict mean = 473k

Majority class vs. Average value

The Overfitting Problem

SHALLOW TREE (depth=3)

Smooth decision boundaries

Train Acc: 88%

Test Acc: 85%

✓ GOOD

DEEP TREE (depth=15)

Jagged, memorizes training data

Train Acc: 100%

Test Acc: 67%

✗ OVERFIT

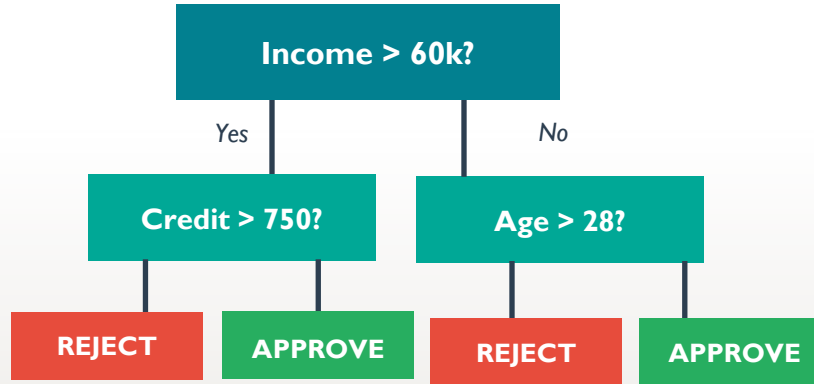
Single trees are too confident - they memorize noise

Exercise Time!

Trace through the trees



Classification Exercise



Predict these:

1. Income=55k, Age=30, Credit=720 → ?
2. Income=75k, Age=25, Credit=730 → ?
3. Income=85k, Age=35, Credit=780 → ?

Classification Solutions

1. Income=55k, Age=30, Credit=720

Income ≤ 60k → Age > 28? → Yes → APPROVE

APPROVE

2. Income=75k, Age=25, Credit=730

Income > 60k → Credit ≤ 750? → Yes → APPROVE

APPROVE

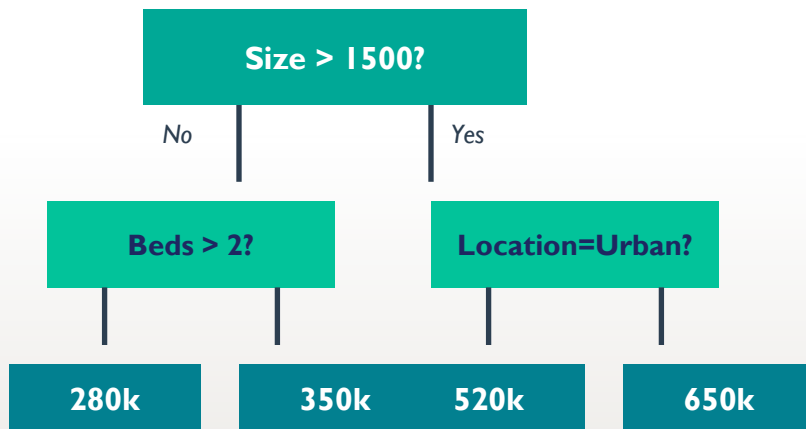
3. Income=85k, Age=35, Credit=780

Income > 60k → Credit > 750? → Yes → APPROVE

APPROVE

Follow the path from root to leaf!

Regression Exercise



Predict prices:

1. Size=1200, Beds=2, Location=Suburban → ?
2. Size=1800, Beds=3, Location=Urban → ?

Notice: regression tree returns values, not classes

Regression Solutions

1. Size=1200, Beds=2, Location=Suburban

Size ≤ 1500 → Beds ≤ 2? → Yes

280k

2. Size=1800, Beds=3, Location=Urban

Size > 1500 → Location = Urban? → Yes

650k

Leaf nodes store mean values from training samples

When to Use Trees

Classification Trees

- ✓ Binary/multi-class problems
- ✓ Class imbalance (adjust weights)
- ✓ Feature importance

Regression Trees

- ✓ Continuous target prediction
- ✓ Non-linear relationships
- ✓ Robust to outliers

Trees Excel At:

- Tabular data (not images/text)
- Mixed feature types (categorical + numerical)
- Non-linear relationships
- Interpretability needs

Questions?

Thank you!